

Calculation Sheet (PDF release)

VBV-T8R1NOFORCE-CALC-01 | case t8_r1_noforce | 2026-08-25 | contract-caliber energy + mass + retention + safety (all from bda tool outputs)

1. Energy calculation (calc-energy, contract caliber)

Metric	Value	Unit	Source
Discharge capacity (1C)	5.2725	Ah	r6_v9_1c_dfn.json
Energy (1C contract)	19.4147	Wh	r6_v9_energy.json
Gravimetric energy density	497.968	Wh/kg	r6_v9_energy.json
Volumetric energy density	854.72	Wh/L	r6_v9_energy.json
Cell mass (contract)	0.0389878	kg	r6_v9_energy.json
Electrode area	0.122915	m2	r6_v9_energy.json
Stack thickness	1.848e-4	m	r6_v9_energy.json
Midpoint voltage	3.9335	V	r6_v9_energy.json
DC resistance (formulation)	0.0007275	ohm	r6_v9_energy.json
Power density (formulation)	150480.7	W/kg	r6_v9_energy.json

2. Mass budget (g/cell = layer_kg_m2 x area)

Layer	kg/m2	g/cell
Positive electrode active material (NMC811)	0.147964	18.1870
Negative electrode active material (graphite)	0.091765	11.2793
Positive current collector (Al, 8 um)	0.021600	2.6550
Negative current collector (Cu, 6 um)	0.053760	6.6079
Separator (10 um)	0.002104	0.2586
TOTAL (contract)	0.317193	38.9878
Electrolyte (supplementary)	-	9.552

3. 5C capacity retention

Quantity	Value	Unit	Source
1C discharge capacity (25 C)	5.2725	Ah	r6_v9_1c_dfn.json
5C discharge capacity (25 C)	5.2178	Ah	r6_v9_5c_dfn.json
Capacity retention @5C	0.989615	-	r6_v9_retention.json
Criterion (entry 0)	0.90	-	log.jsonl entry 0
Verdict	PASS	-	0.9896 >= 0.90
5C discharge T_max	313.64	K	r6_v9_5c_dfn.json (<= 333.15 K)

4. Safety exam (4C charge at 45 C, lumped thermal + plating module)

Quantity	Value	Unit	Determination
Anode potential minimum	0.0063475	V	plated = false -> PASS (margin +6.3 mV)
Lithium plating flag	false	-	PASS (criterion: plated = false)
T_max 4C charge	326.81	K	PASS (<= 333.15 K red line)
T_max 5C discharge	313.64	K	PASS (<= 333.15 K red line)
T_max red line (entry 0)	333.15	K	protocol default (task silent)

5. Notes and assumptions

* Electrolyte excluded from contract mass (calc-energy contract caliber; parameter set carries no electrolyte density).

- * Electrolyte transport $\kappa = 1.4 \text{ S/m}$, $D = 3.7\text{e-}10 \text{ m}^2/\text{s}$: literature values (Nyman 2008) set as formulation levers, recorded in R6 propose.
- * Electrolyte density 1.2 g/cm^3 : literature value for BOM row / datasheet mass incl. electrolyte (48.54 g).
- * Binder/additive fractions: literature defaults, excluded from contract mass.
- * N/P = 0.641 capacity-density basis (neg 49.18 Ah/m² vs pos 76.72 Ah/m²) - negative-limited by design.
- * Formation recommendation: 0.1C CC to 4.2 V, 25 C, 2 cycles (design rec.; production tuning required).
- * Not simulated: cycle life, low-T retention, abuse (nail/crush/overcharge), self-discharge - recorded N/A in DVPR.
- * R5 V8 5C discharge IDA_CONV_FAIL recorded verbatim in log.jsonl; superseded by V9 design.